

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)			Indicative content molar mass is 72 g mol^{-1} of which 50.0% is carbon $\Rightarrow$ each molecule of compound G must contain 3 carbon atoms this leaves a mass of 36 $\Rightarrow$ must be 2 oxygen atoms and 4 hydrogen atoms as 1 oxygen atom and 20 hydrogen atoms is not feasible molecular formula is $\text{C}_3\text{H}_4\text{O}_2$ it reacts with Tollens reagent $\Rightarrow$ must be an aldehyde / have the CHO group it gives a yellow solid with alkaline iodine $\Rightarrow$ must have a $\text{CH}_3-\text{C}(=\text{O})$ group or a $\text{CH}_3-\text{C}(\text{OH})(\text{H})$ group only 2 hydrogen environments, CH_3 group (at ~ 2.2) and CHO group (at ~ 9.4) $\Rightarrow$ suggests $\text{CH}_3-\text{C}(=\text{O})-\text{C}(=\text{O})\text{H}$ reduction of this compound gives a diol with molar mass 76 g mol^{-1} $\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{OH} \\ \quad \\ \text{OH} \quad \text{H} \end{array} \Rightarrow \text{confirms above structure}$	2	2	2	6		2